
ASSA ABLOY is the global leader in door opening solutions, dedicated to satisfying end-user needs for security, safety and convenience
We Make Possible
ASSA ABLOY
Sectional doors

ASSA ABLOY overhead sectional doors

OH1042P



Panel

OH1042F



Frame

ASSA ABLOY OH1042F



ASSA ABLOY overhead sectional doors portfolio

OH1042P



OH1042F



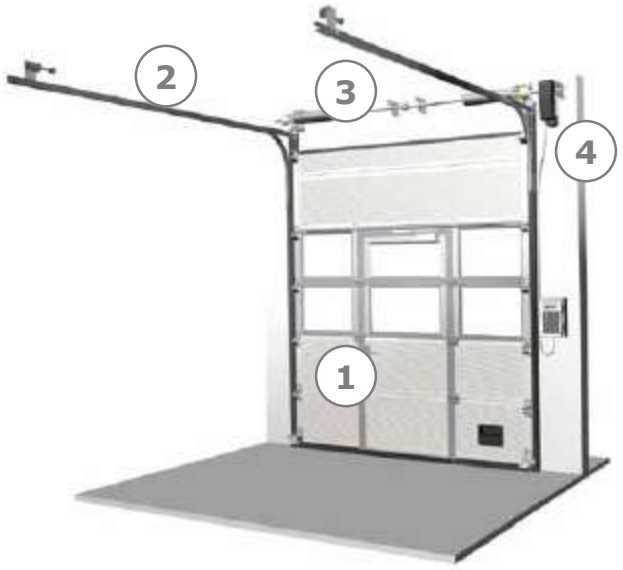
Combination
OH1042P & OH1042F



Design

ASSA ABLOY overhead sectional door - Parts

The door can be divided in 4 groups



1. Door leaf
2. Tracks
3. Spring package
4. Operation (Optional)

ASSA ABLOY OH1042P – NIHVA Edition



ASSA ABLY OH1042P
NIHVA specification

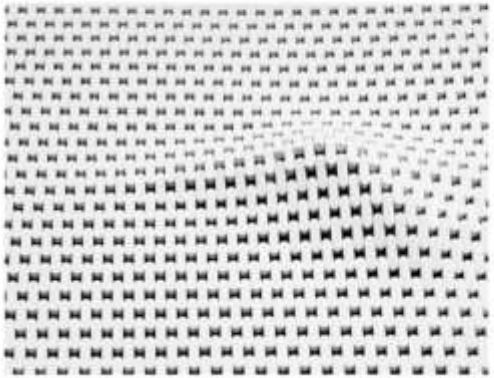
- Min. size: 2200 x 2500 mm (W x H)
- Max. size: 4000 x 4000 mm (W x H)
- Panel: 42 mm Square Grid, steel sheet
- Filling: FC-free polyurethane
- Weight: Steel: 13 kg/m^2
- Colour outside/inside: RAL 9002 Grey White
- Track type: Standard Lift (SL)
- Lock: Lock bolt
- Manoeuvring: Lift/step handle and Pull down rope
- Sealing: Top, bottom and side sealing

Crawford OH1042P technical facts and features, Optional

- Track types: HL (1000 mm), VL
- Windows: DARP
- Painting: Acc. to RAL scale
- Operation: Chain hoists / Electrical operation

Selected panel design







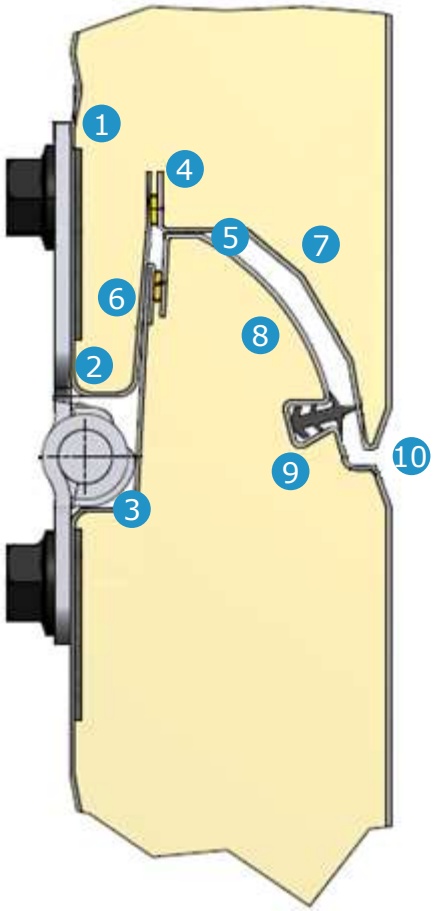


ASSA ABLOY OH1042P door leaf

- The Crawford OH1042P overhead sectional door leaf has 545mm high sections, connected together with hinges.
- Height adjustment will take place in the top section
- The outer hinges of each section have rollers that run in the tracks.
- The horizontal sections are insulated panels, filled with FC-free polyurethane.



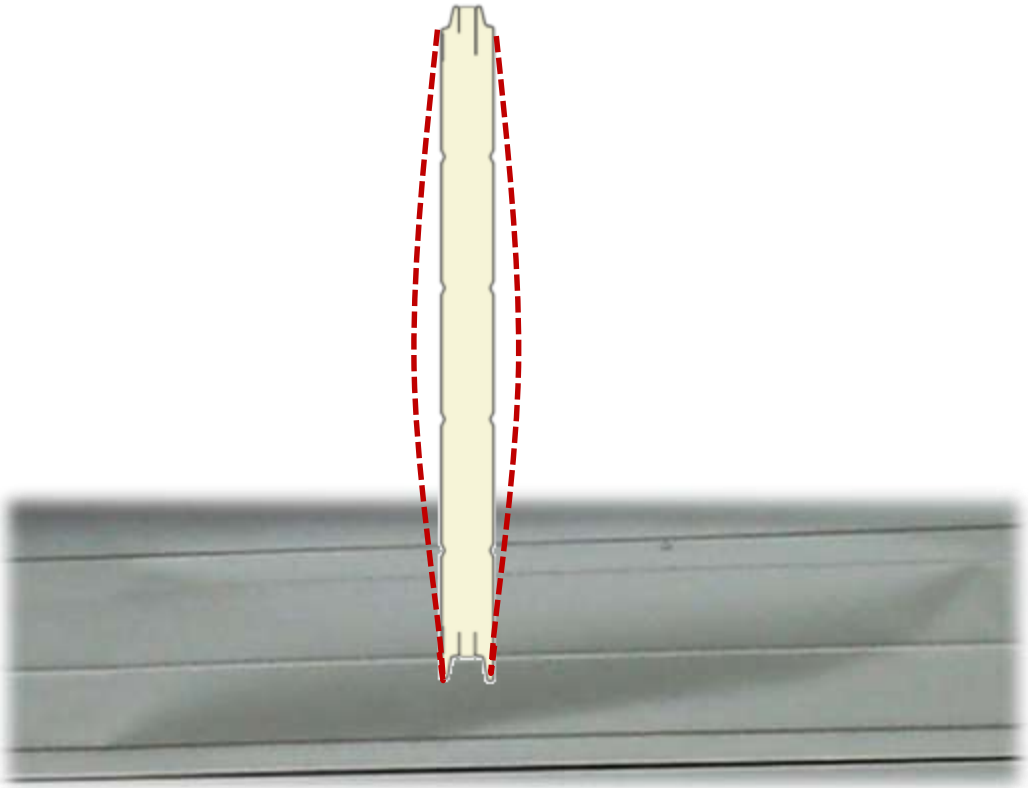
Section joint



Nbr.	Text
1	Steel reinforcement
2	Hinge line gives lower nose
3	Hinge with cams to avoid rubbing against plate
4	Hotmelt to break the cold bridge
5	Panels supporting each other to create stability
6	Sheet folded 180 degrees to avoid rust and to create stability
7	Not round but folded for higher stability
8	Round form with centre in hinge line (finger pinch protection)
9	Sealing invisible in groove
10	Kinked edge to create smooth transition

What happens with poor load distribution

De-lamination caused by poor load distribution



Cracks caused by poor load distribution

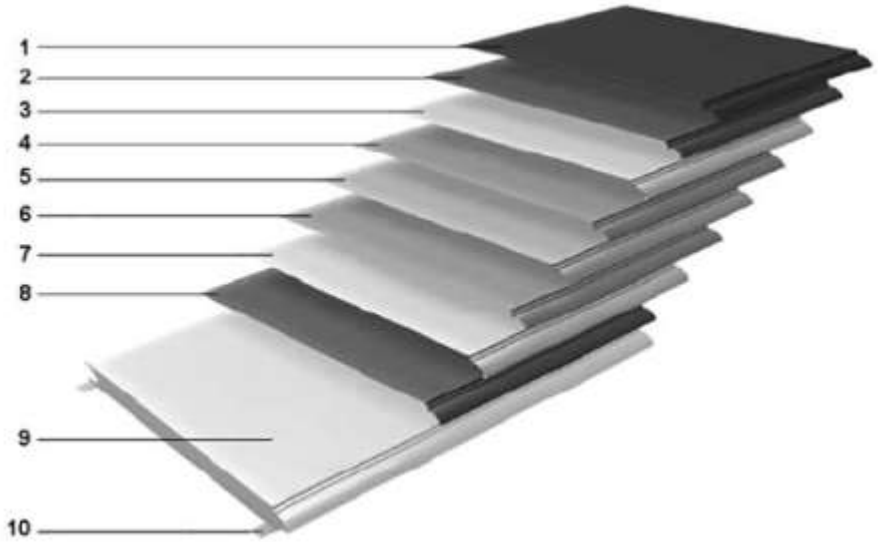




Material

The surface of the door leaf panels is characterized by the square grid steel sheet.

1. Polyester coating
2. Primer
3. Chromate layer
4. Zinc coating
5. Steel sheet
6. Zinc coating
7. Chromate layer
8. Primer
9. FC-free polyurethane
10. Reinforcement strips

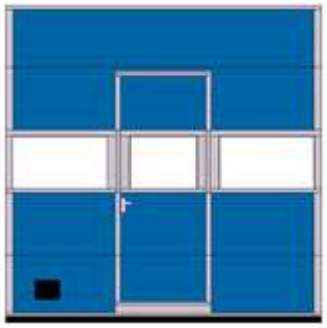


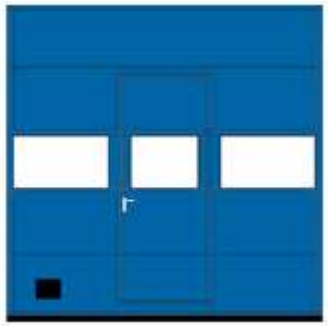
Optional colours

- The door leaf can be factory painted in any RAL, NCS or BS colour, outside, inside or both sides

- The painting can be applied to the panel or to the complete door leaf, including frames and strips.







Seals

The door is equipped with well designed sealing's on all sides which gives the door its excellent sealing abilities.

Bottom seal



Installed on the bottom edge of the bottom panel, to act as a barrier as well as a shock absorber. The flexible rubber material and the O-shape provides continuous pressure on the floor, ensuring maximum sealing.

Top seal



Installed on the top panel, to seal the gap between the panel and the wall. The flexible rubber material provides continuous pressure on the top wall, ensuring maximum sealing.

Side seal



Installed on the track set to close the gap between the tracks and the door leaf. The flexible rubber material provides continuous pressure on the door leaf, while dodging irregularities, ensuring maximum sealing.

Handle

- For manual operation, every ASSA ABLOY OH1042P overhead sectional door is provided with a solid, easy to grip and step–on handle, finished with the Crawford logo.



DARP windows

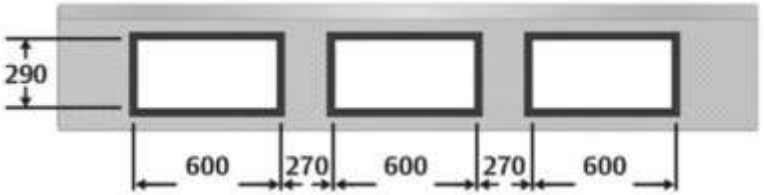
DARP

- **D**ouble **A**crylic (3 + 2 mm) **R**ectangular, in **P**olycarbonate frame
- Light opening: 600 x 290 mm
- Window frame in black polycarbonate

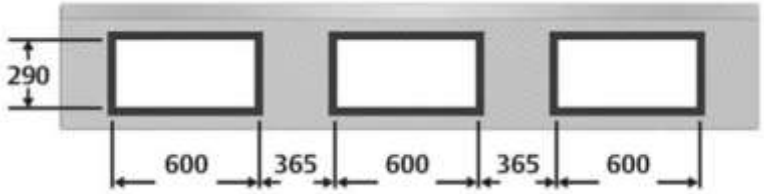


DLW Without Pass door	DLW With Pass door	No. of Windows
2050 - 2134	2050 – 2234	1
2135 – 2999	2235 – 3199	2
3000 – 3864	3200 – 4164	3
3865 – 4729	4165 – 5129	4
4730 - 5594	5130 – 6050	5
5595 - 6459		6
6460 - 7324		7
7325 - 8000		8

Without Pass door



With Pass door



Lock bolt

ASSA ABLOY 1042P is equipped with a Lock bolt.



The Lock bolt locks the door from the inside, without the use of a key.

The Lock bolt is not visible from the outside.

Safety

The balancing system supports heavy forces.
In case of a spring or cable break, its counterforce is lost.
The door is therefore equipped with two safety devices that can block downward door movement:

- Spring Break Device (standard)
  - In the event of a spring break, the sudden drop force activates the Spring Break Device.



- Cable Break Device (optional)
  - In the event of a cable break the cable looses its tension. The activated Cable Break Device clamps onto the track.



ASSA ABLOY OH1042P, Performance

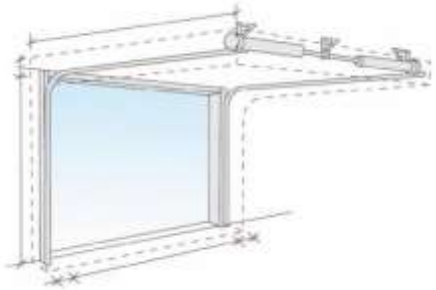
Opening/closing speed:	CDM9: 0,25 m/s
Life time expectations:	Door: 100.000 door cycles Springs: 20.000 door cycles
Wind load EN12424 *)	Class 3 (= 4250 mm DLW) Class 2 (> 4250 mm DLW)
Thermal transmittance EN12428	1,1 W/(m2.k) Steel door 1,25 W/(m2.k) With windows/pass door (4000 x 4000 mm)
Water penetration EN12425	Class 3 (4000 x 4000 mm)
Air permeability EN12426	Class 3 (4000 x 4000 mm, with pass door class 2)

*) Higher wind load classification on request

Hardware - Types

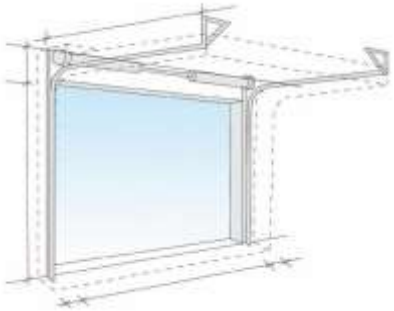
Low Lift (Option)

(head room < 500mm)



Standard Lift (Std.)

(head room 500 - 1000mm)



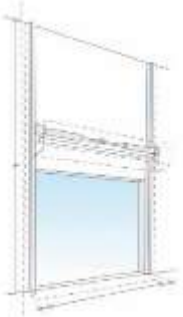
High Lift (Option)

(head room > 1000mm)

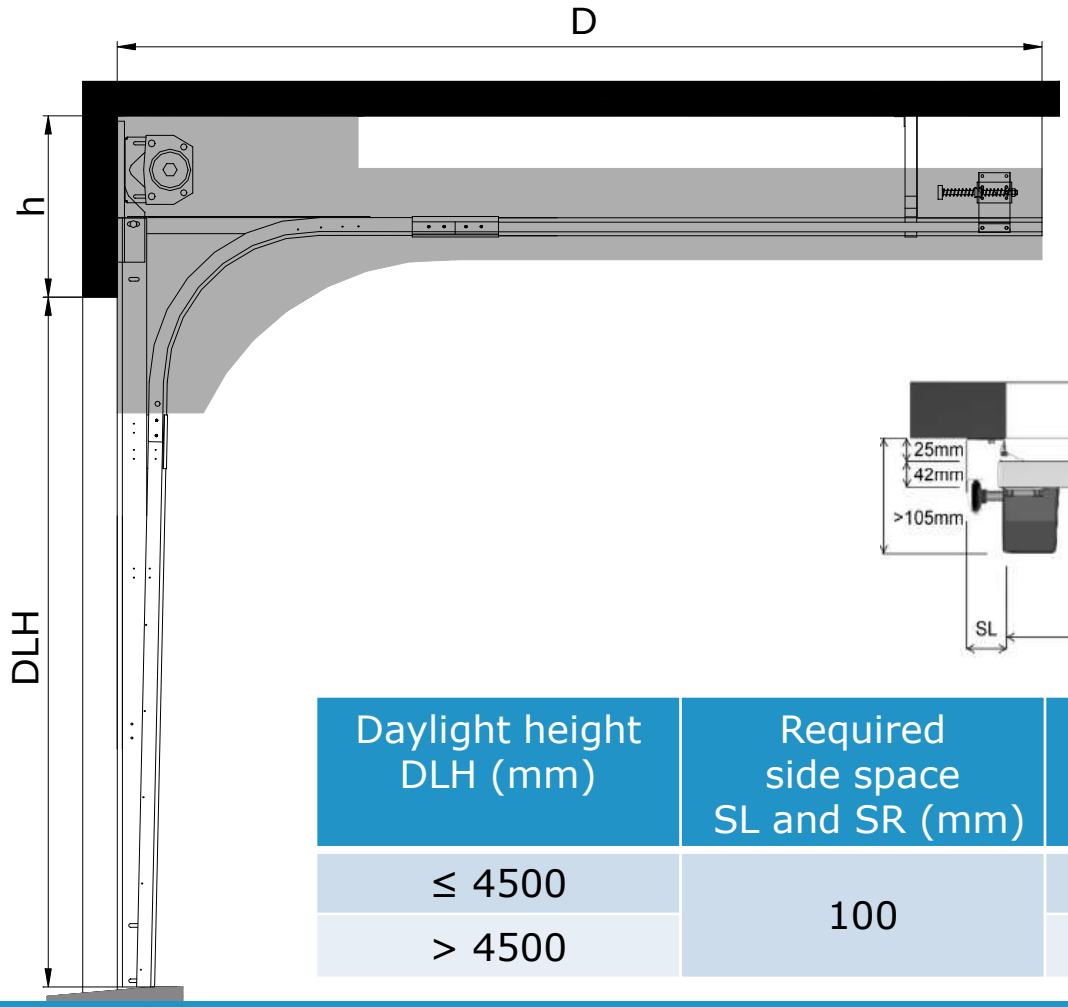


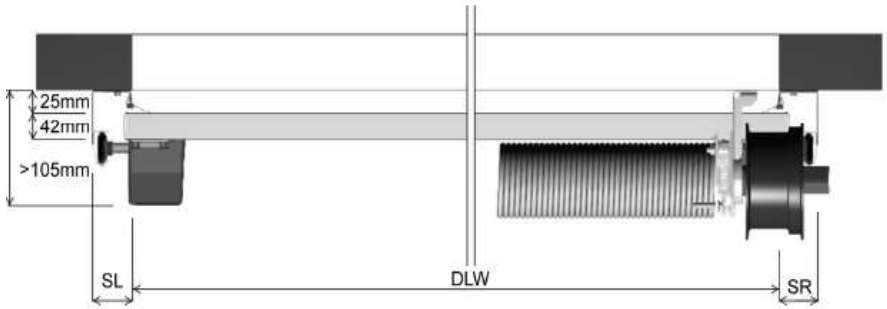
Vertical Lift (Option)

(head room > 2 x height)



Standard Lift (SL)

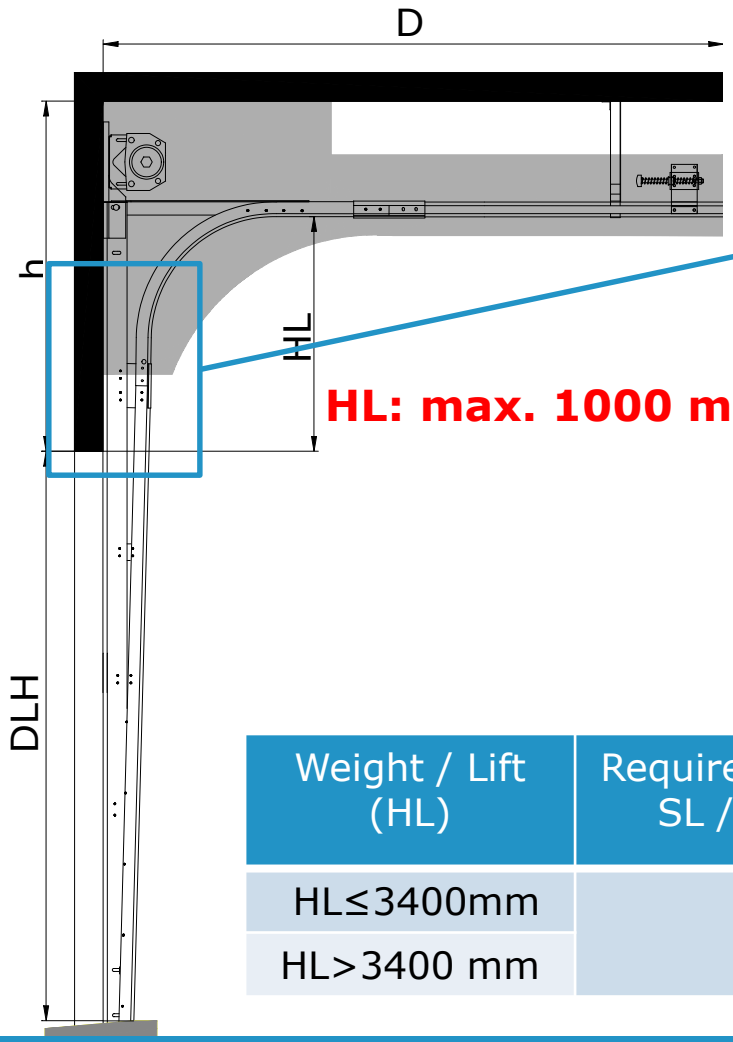




Daylight height DLH (mm)	Required side space SL and SR (mm)	Required excess height h (mm)	Required depth D (mm)
≤ 4500	100	485	DLH + 600
> 4500		510	

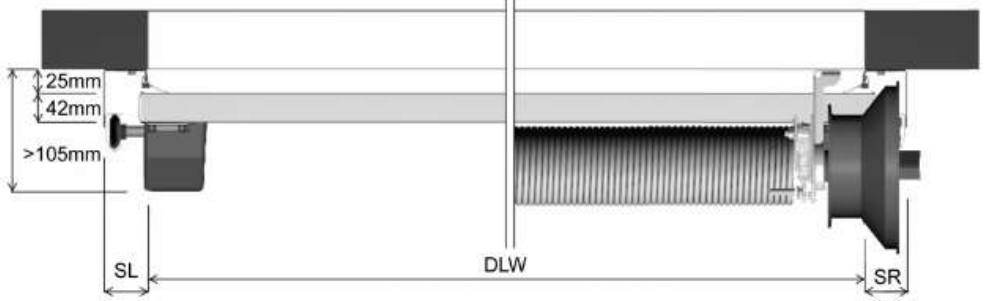
High Lift (HL)

HL mounted as VL on a beam		
Optional if:	HL > 1500 mm	DLW < 6000 mm



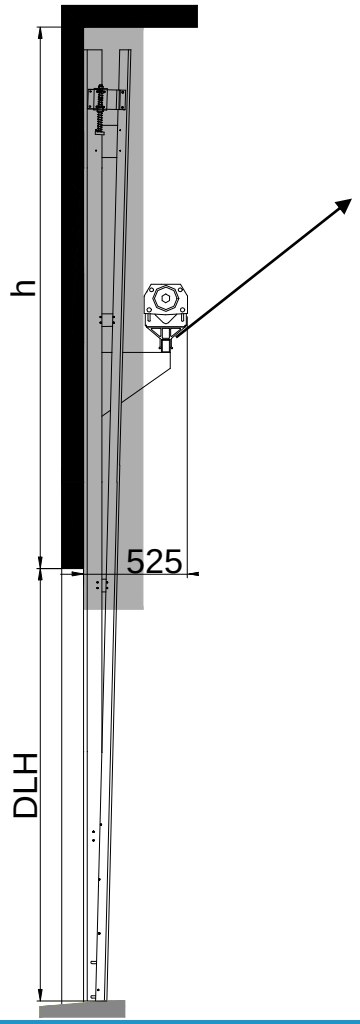
HL: max. 1000 mm



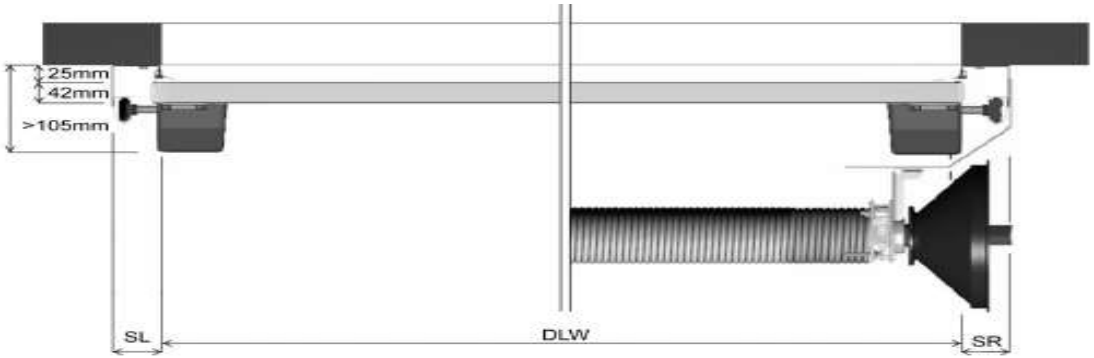


Weight / Lift (HL)	Required side space SL / SR (mm)	Required excess height h (mm)	Required depth D (mm)
HL≤3400mm	100	HL + 320	DLH - HL + 800
HL>3400 mm		HL + 370	

Vertical Lift (VL)

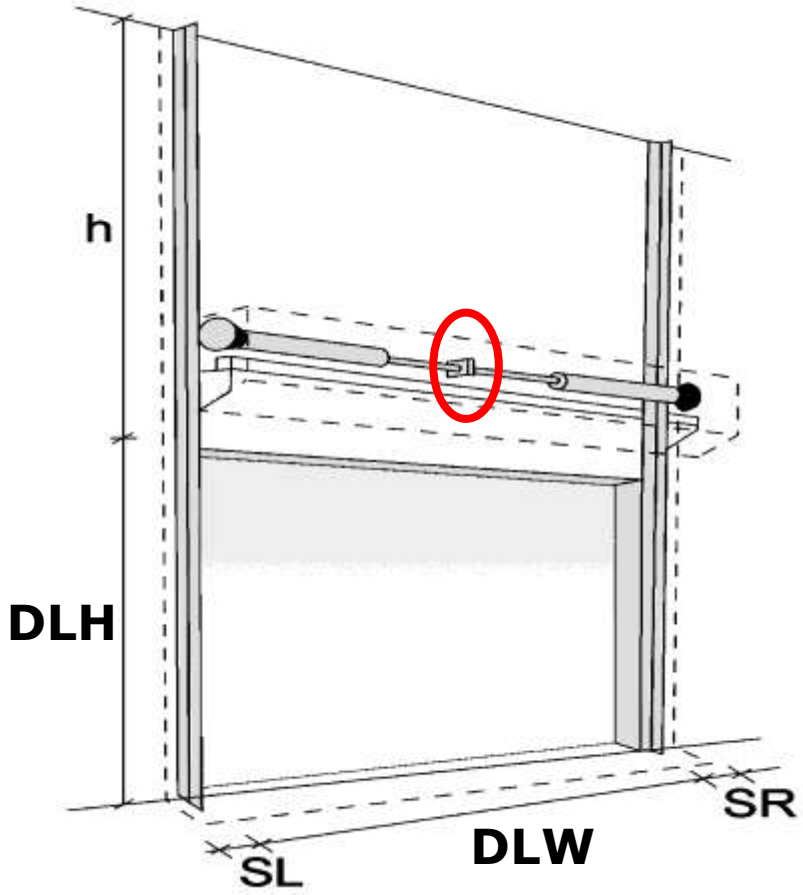


DLW ≤3000 mm no beam installed
DLW >3000 mm installed beam to support the balancing system



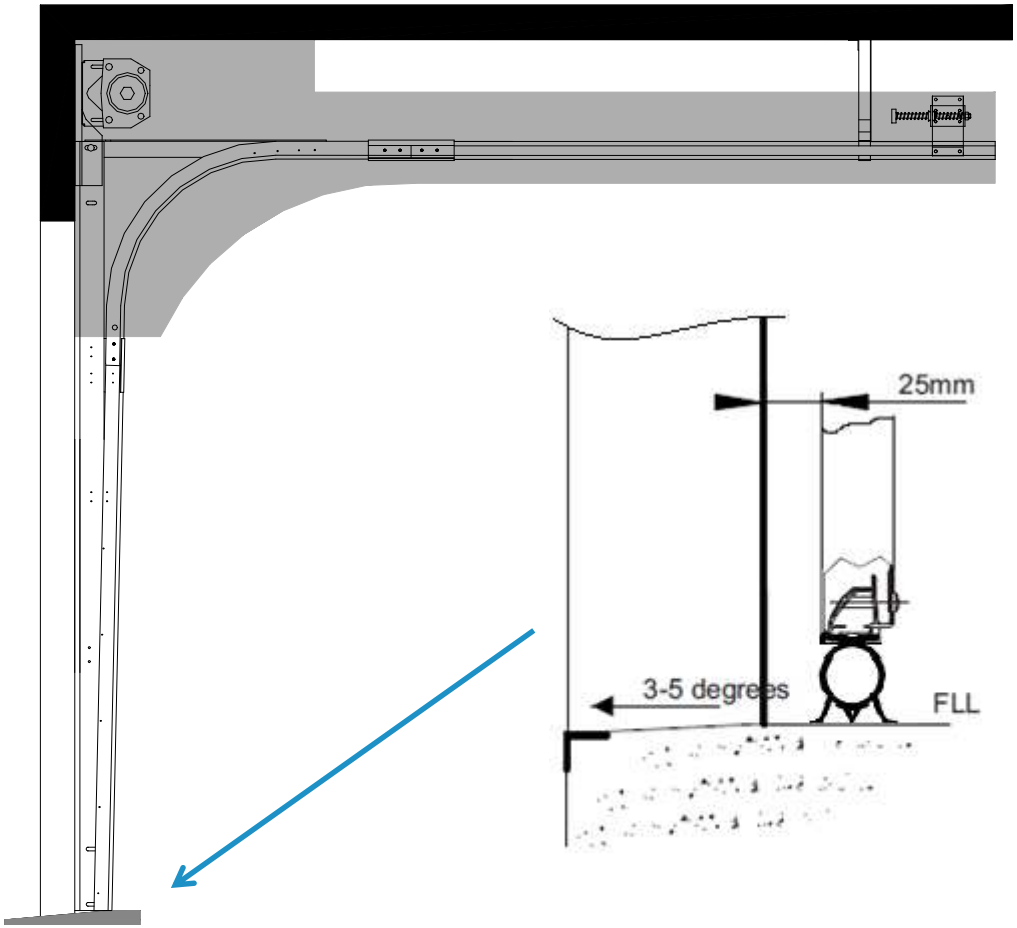
Daylight height DLH (mm)	Required side space SL / SR (mm)	Required excess height h (mm)	Required depth D (mm)
≤ 5500	100	DH + 400	525

Center bearing



If DLW>3000 center bearing is needed!

Recommended floor detail



Recommended 3-5 degrees slope on the floor

Falling from inside out

How to manoeuvre the door?

Operation



Pull-down rope

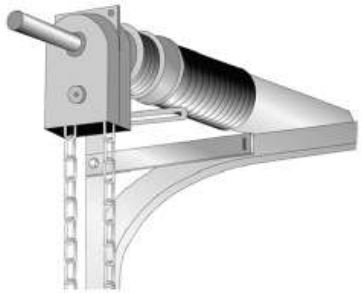
- The Crawford overhead sectional door can be operated manually with a pull-down rope.
- The pull-down rope is directly connected to the door leaf.



Chain hoist

For heavier doors, a chain hoist allows easier door operation. There are three types of chain hoist:

T-hoist



Geared (ratio 1:4) chain transmission directly connected to the shaft.

Recommended for doors up to 250 kg.

For all shaft types.

Why an operator?

5 Major Advantages

- Reduced opening time give less energy losses
- Reduced passage time give efficiency
- Controlled opening and closing give less wear and tear on the door
- Less draughts and a better indoor climate give a good environment
- Lifting a door frequently by hand increase the risk of bad health

Lower total cost for the end user!

Selection guidelines for operation type

Door size m^2	1-5 openings/day	5-10 openings/day	10-15 openings/day	> 25 openings/day
0-10	Manual / operator	Manual / operator	Operator	Operator + automation
10-20	Manual / operator	Operator	Operator + automation	Operator + automation
> 20	Geared hoist / operator	Operator	Operator + automation	Operator + automation

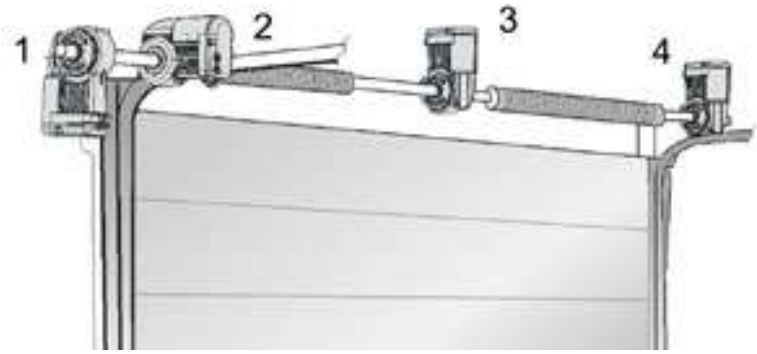
Operator CDM9 - Door control systems

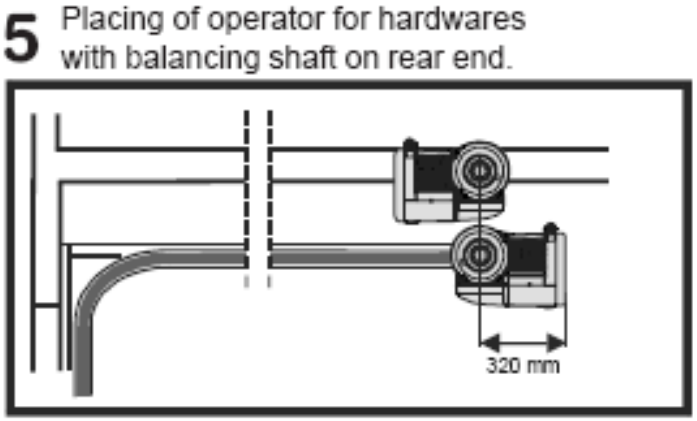


- The CDM9 operator is a combination of the CDM9 operator and a 900-series Door control system.
- This system is available for doors up to 800 kg.
- One main part of the system is the operator: an electric motor which drives the balancing shaft with the cable drums and torsion springs. It can be retrofitted to an already installed door. The CDM9 operator is mounted directly on the balancing shaft and does not require any special wall reinforcement.
- Key features:
  - Smooth and silent
  - Soft start and stop
  - Fits all track types and shafts
  - Life time: 84.000 - 300.000 door cycles *)
    (depending on weight and temp.) e.g.:
    –temp. 0 to 40 °C/weight 250 kg = 300.000 cycles
    –temp. -20 to +60 °C/weight 400 kg = 84.000 cycles

*) For door weights between 650kg and 800kg the endurance becomes 50.000 cycles

Crawford CDM9 Installation Space





Location	Space requirement (in mm) beyond the requirements for a manually operated door.	
	Side room (mm)	Head room (mm)
1 Left/right	200	0
2 Left/right	200	0
3 Centre	0	320
4 Left/right	200	320
5 Left/right	200	0
5 Centre	0	0

Door control systems – Selection guidelines

Functions included	920	930	950
			
Open (by impulse)	X	X	X
Open (hold to run)			
Stop	X	X	X
Close (by impulse)		X	X
Close (hold to run)	X		
Safety edge		X	X
Open function		X	X
One button function			X
Display (diagnostics)			X
Service indicator			X

Door Control Systems 900



Crawford 930 Door control system

The Crawford 930 Door control system is a basic control unit that has impulse up and down functions with supervised safety edge.

An extra set of upgrade features, such as external control, that can be installed during or after installation, is available.

This control unit is the more advanced solution for door openings that are frequently used by pedestrians and forklift trucks, because of its automated opening and closing function.



Crawford 950 Door control system

The Crawford 950 Door control system is the most advanced control unit that is prepared for one or more physical upgrades from the entire range of automation systems. An automation system allows door operation by sensors or remote control.

This control unit contains a 3-digit diagnostics display that allows efficient troubleshooting and displays the number of door cycles. Together with the service indicator, this extra

feature allows advanced maintenance planning to users where the door is an essential element of internal logistics.

CDM9 Options

Functions optional		930	950
			
Complete kits			
Automation D-kits			X
Basic control functions			
Interlocking			X
Reduced opening			X
External control functions			
External pushb. box		X	X
Pull-rope switch		X	X
Remote control open/stop/close		X	X
Remote control 1-button function			X

Functions optional		930	950
			
Automatic control functions			
Automatic closing		X	X
Photocell open door			X
Safety functions			
Safety photocell (1 or 2)		X	X
French safety logic			X
Additional functions			
UPS Battery backup		X	X
Relay box			X

ACCESS AND AUTOMATION (1/2)

Interlocking



Developed for climate control or safety; If door A is open, door B cannot be opened. If door B is open, door A cannot be opened.
An interlocked door can remember an up-command, if selected via a micro switch.
Circuit card Installed in control unit.

External push button box



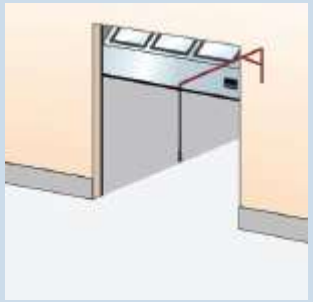
An extra control box is installed outside the building or inside close to the door if the main control unit needs to be installed away from the door opening.
Installed on the inside or outside wall beside the door.

Reduced opening



When it is unnecessary or undesirable to fully open a door, an additional switch can be used to open the door to a preprogrammed reduced opening position.
Pre-fitted micro-switch to be activated in control unit.

Pull-rope switch



A pull-rope switch above the door opening can be operated from e.g. a forklift truck. Pulling the rope opens a closed door or closes an opened door.
Installed on the inside construction above the door.

ACCESS AND AUTOMATION (2/2)

Remote control



A hand-held radio transmitter allows door operation from a vehicle or any position within 50- 100 meters from the receiver and aerial at the door. For closing, the door can be provided with a photocell beam.
Receiver installed in control unit, antenna installed on the wall beside the door.

Radar



An infrared sensor above the door detects an object (person, vehicle) within a specified distance from the door and opens the door automatically. This is an ideal solution for frequent vehicle or personal traffic. Often combined with automatic closing.
Installed on the inside or outside wall above the door.

Magnetic loop



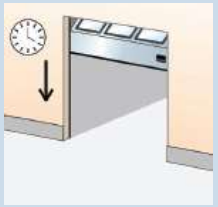
A sensor in the floor detects a metal object (usually forklift trucks, pallet trucks) and opens the door automatically. This is an ideal solution for frequent vehicle traffic. Installed on the outside, inside or both sides of the door in the floor.

Photocell open door



A set of photocells on pillars, on each side of the door. When a person or vehicle passes between the photocells, the beam is interrupted and the door opens. Photocells installed on pillars, away from the door.

Automatic closing



A programmable timer that closes the door after a specified time, counted from either the fully open position and/or from passing through the photocell beam. Adjustable micro switches in control unit.

SAFETY FUNCTIONS (1/2)

Safety edge 1)



As a standard, all doors that have the impulse-close function or any form of automated closing, are equipped with a safety edge. The pneumatic sensor in the bottom seal detects any obstruction under a closing door and reverses the door.
Installed in the bottom seal.

Safety photocells 2-channel



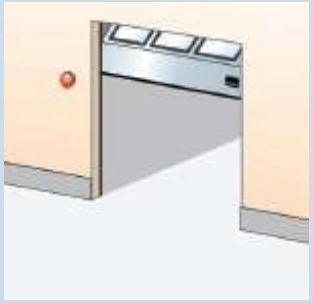
Two sets of photocell transmitter and receiver are installed in the door opening. If one or both photocell beams are interrupted during closing, the door will stop in less than 30mm and reverse to the fully open position.
Installed in the door opening.

Safety photocells 1-channel



A set of a photocell transmitter and receiver is installed in the door opening. If the photocell beam is interrupted during closing, the door will stop in less than 30mm and reverse to the fully open position.
Installed in the door opening.

Warning lights - Red



Two red warning lights giving information on the current door behavior. Flashing light before or during door movement.
Optional: Continuous light before and during door movement.
Installed on the inside and outside wall beside the door.

1) Retractable Photocells see separate page

SAFETY FUNCTIONS (2/2)

Warning lights - Green



One or two green warning lights indicating the open position of the door by continuous light signal.
Installed on the inside and/or outside wall beside the door.

UPS battery backup



When mains failure cannot be permitted or an increased risk of mains failure is predicted, the UPS battery backup system can be installed to store enough energy for 10 door cycles.
Installed on the inside wall beside the door.

Traffic lights - Red & Green



If traffic through a door needs to be directed; two red and two green traffic lights can be installed to indicate traffic direction. From the side where a vehicle is first detected to approach the door, the green traffic light comes on. The opposing side shows a red traffic light. Traffic from this direction must give way to the other. Usually installed in e.g. parking garages.
Installed on the inside and outside wall beside the door.

Relay box



A sealed connection box makes it possible to safely connect external high-voltage equipment.

Impressions







www.nihva.com